Responsible use of Large Language Models for Research: Data collection and management

A practical session on LLM utility for preparing data

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Course website: https://bma-vandijk.github.io/RULLMR/



Programme

1. Scope of this session
2. Workflow design and data management
3. Possible approaches to homework assignment 1
4. Finish and compare your homework in teams

Learning goals

After this session you should be able to:

- Explicate and design a workflow for LLM-driven data collection and management
- Practice with and explicate performance differences in using local and cloud LLMs for data collection

1. Scope of this session

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We focus on common preprocessing and extraction tasks while keeping good DMP practices:

- Cleaning and standardizing noisy tables;
- Transforming unstructured text to a structured dataset;

Note that this session will be a bit more practical and technical

1.1 Assignment 1 recap: Cleaning and standardizing tables

noisy_ehr.csv

- What issues did you find with this homework table?

patient_id,appointment_date,birth_date,age,referral_reason,gender,ethnicity,insurer
PT-10482,04/02/2026,03/14/1968,58,HTN f/u,M,White,BCBS
pt-10482,2 Apr 2026,14-03-1968,58 years,hypertension follow-up,Male,Caucasian,Blue Cross Blue Shield
PT10482,2026/04/02,1968-03-14 00:00:00,57,htn follow up,m,white,Blue Cross
PT 10482,04-02-2026,March 14 1968,58yo,high blood pressure,Man,W,BC/BS
….

1.1 Assignment 1 recap: Cleaning and standardizing tables

noisy_ehr.csv

- 1) many values are inconsistent; 2) some even invalid
- Standardisation (of all columns) and validation are two different tasks, but both based on assumptions that should be explicated somewhere
  - E.g. For a single patient ID, ambiguous dates are left as-is, or a single visit is assumed based on other rows

patient_id,appointment_date,birth_date,age,referral_reason,gender,ethnicity,insurer
PT-10482,04/02/2026,03/14/1968,58,HTN f/u,M,White,BCBS
pt-10482,2 Apr 2026,14-03-1968,58 years,hypertension follow-up,Male,Caucasian,Blue Cross Blue Shield
PT10482,2026/04/02,1968-03-14 00:00:00,57,htn follow up,m,white,Blue Cross
PT 10482,04-02-2026,March 14 1968,58yo,high blood pressure,Man,W,BC/BS
….

1.1 Assignment 1 recap: Cleaning and standardizing tables

Desired result

- ID standardized according to specific format, date according to format + assumption

patient_id,appointment_date,birth_date,age,referral_reason,gender,ethnicity,insurer
PT-10482,04/02/2026,14/03/1968,57,HTN f/u,M,Caucasian,Blue Cross Blue Shield
PT-10482,02/04/2026,14/03/1968,58,hypertension follow-up,M,Caucasian,Blue Cross Blue Shield
PT-10482,04/02/2026,14/03/1968,57,htn follow up,M,Caucasian,Blue Cross Blue Shield
PT-10482,04/02/2026,14/03/1968,57,high blood pressure,M,Caucasian,Blue Cross Blue Shield
….

1.2 Assignment 2 recap: from free text to structured data

homework_10.csv

- **Free text with information scattered throughout**

"case_id","report"
"ED-003","Reden bezoek
Buikpijn volwassene

Anamnese
rvk: retour buikpijnklachten.
Sinds 4 dagen geen def, daarvoor wat dunne ontlasting. Eerder op SEH beoordeeld, echo met gevulde darmlissen, gedacht aan obstipatie.
Laxantia en clisma gebruikt, vooral krampen, weinig resultaat. Ook weinig winden. Minder eetlust, boeren, niet gebraakt. Buik voelt boller. Pijn blijft ondanks eerdere behandeling.

vg
Obstipatie vaker
Buikoperatie in verleden, verdere gegevens niet genoteerd

med
Movicolon
Pantoprazol

Lichamelijk onderzoek
Opgetrokken benen, pijnlijke indruk, ligt liefst stil. Niet acuut ziek.
Abd slank, wat opgezet volgens pte. Spaarzame zachte peristaltiek. Drukpijn LOQ met aanspannen, nierloges geen slagpijn, WT.
rr148/84 p86 sat98% temp36.8

1.2 Assignment 2 recap: from free text to structured data

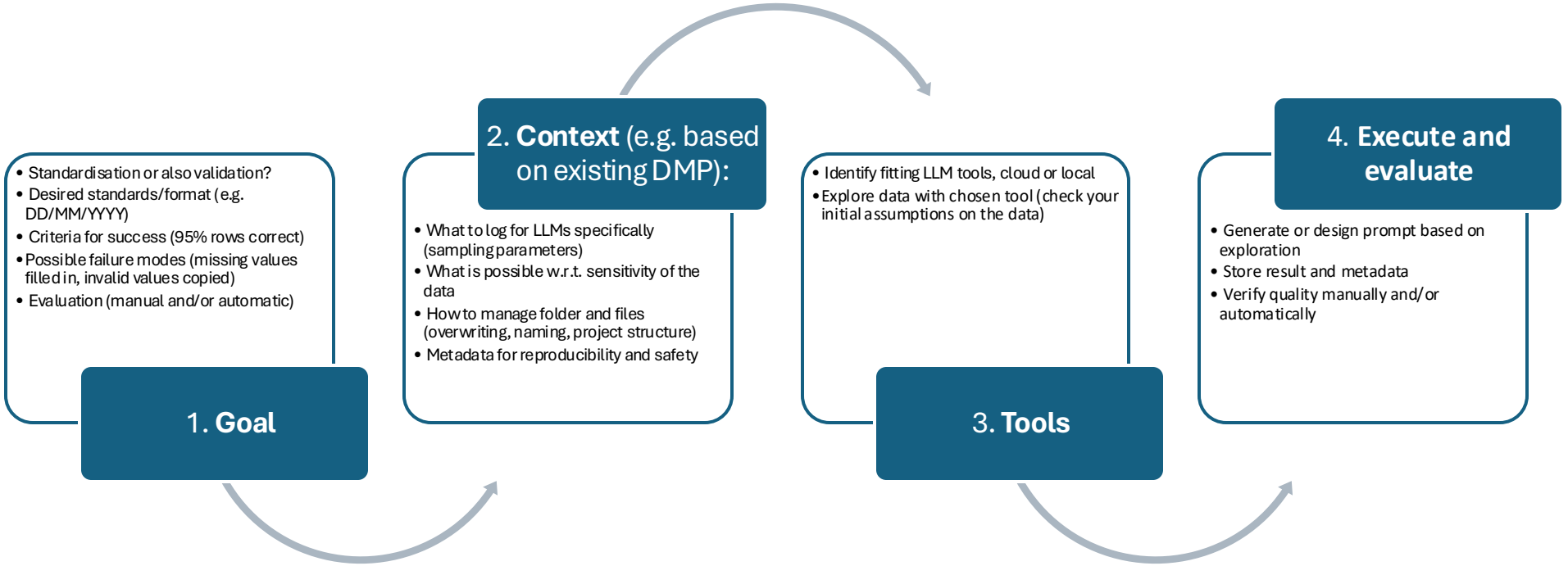
Desired result

```
case_id,reason_for_visit,temperature,BP,saturation,medications
ED-003,Buikpijn bij volwassene; herbezoek wegens aanhoudende buikpijn,36.8,148/84,98%,Movicolon; pantoprazol
```

2. Workflow design and data management

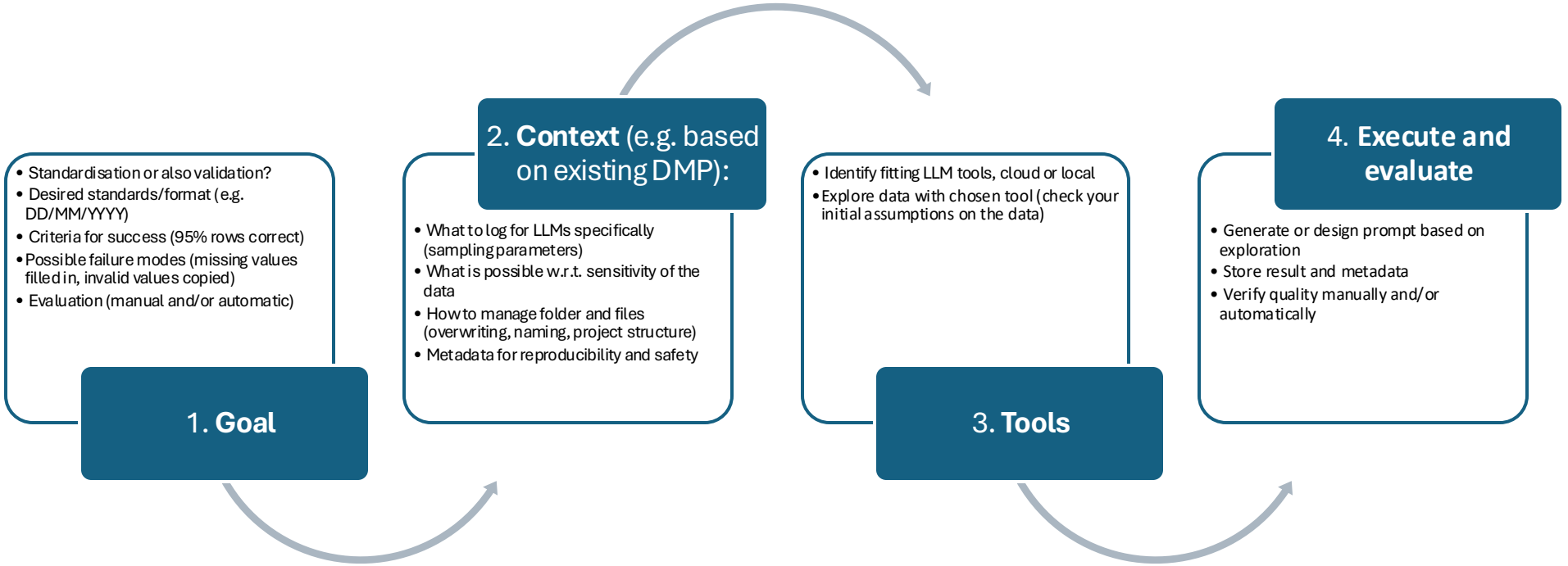
2. Assignment 1: Cleaning and standardizing tables

A possible, simple workflow could involve this:



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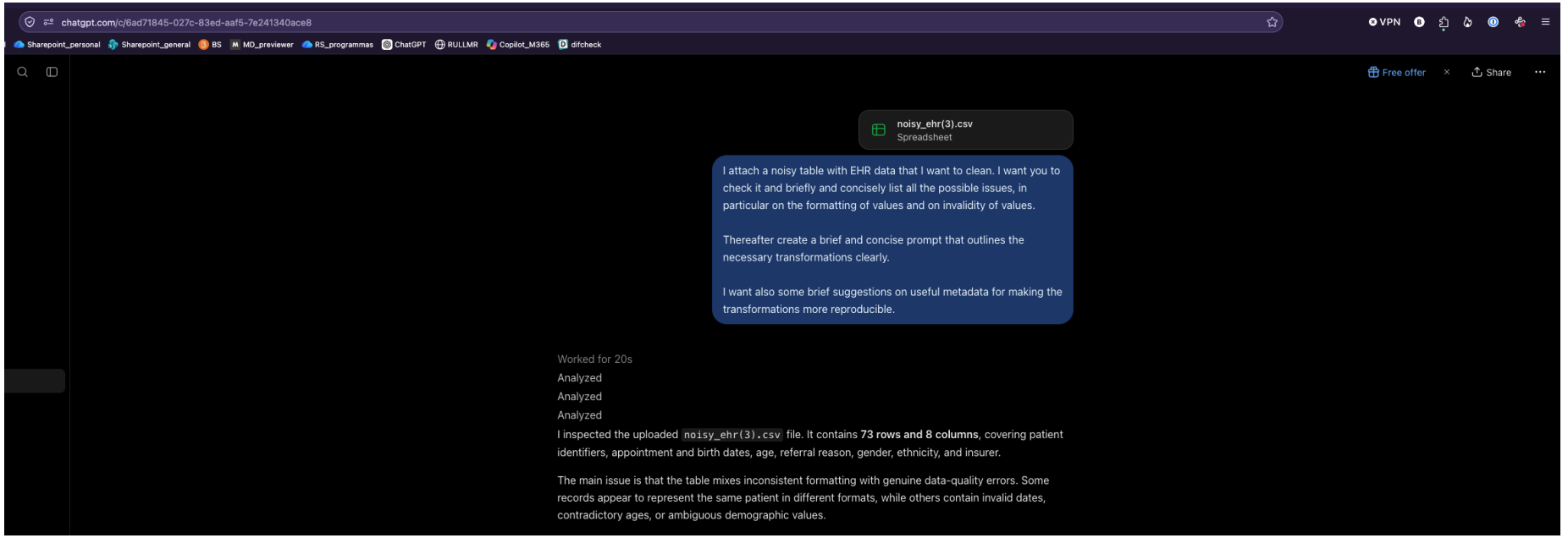
A possible, simple workflow could involve this:



3. Approaches to assignment 1

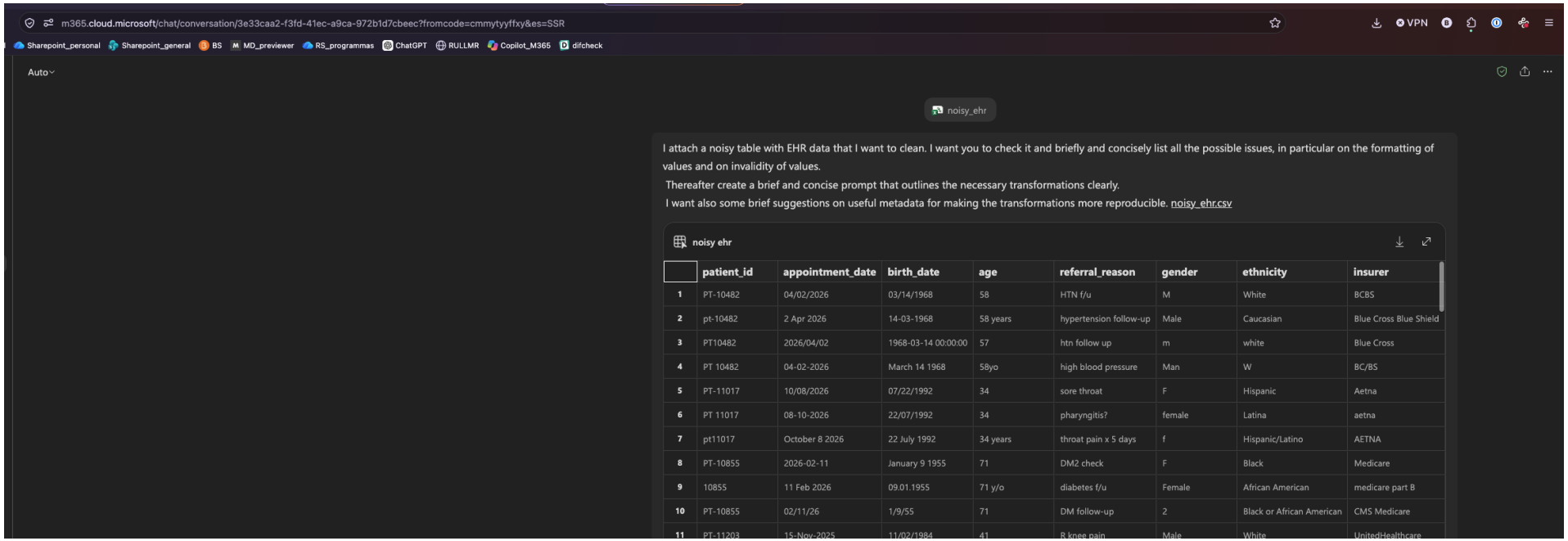
3.1 Assignment 1 with commercial LLMs

- **Explore dataset and prompt generation** with a SOTA commercial model: https://chatgpt.com/share/6ad7198d-c44c-83eb-9272-9652203330a2
- Check whether findings complement/contradict/confirm what you already know
- Some of the info can be used for completing your workflow
- Note the assumptions already made for you



3.2 Assignment 1 with commercial LLMs

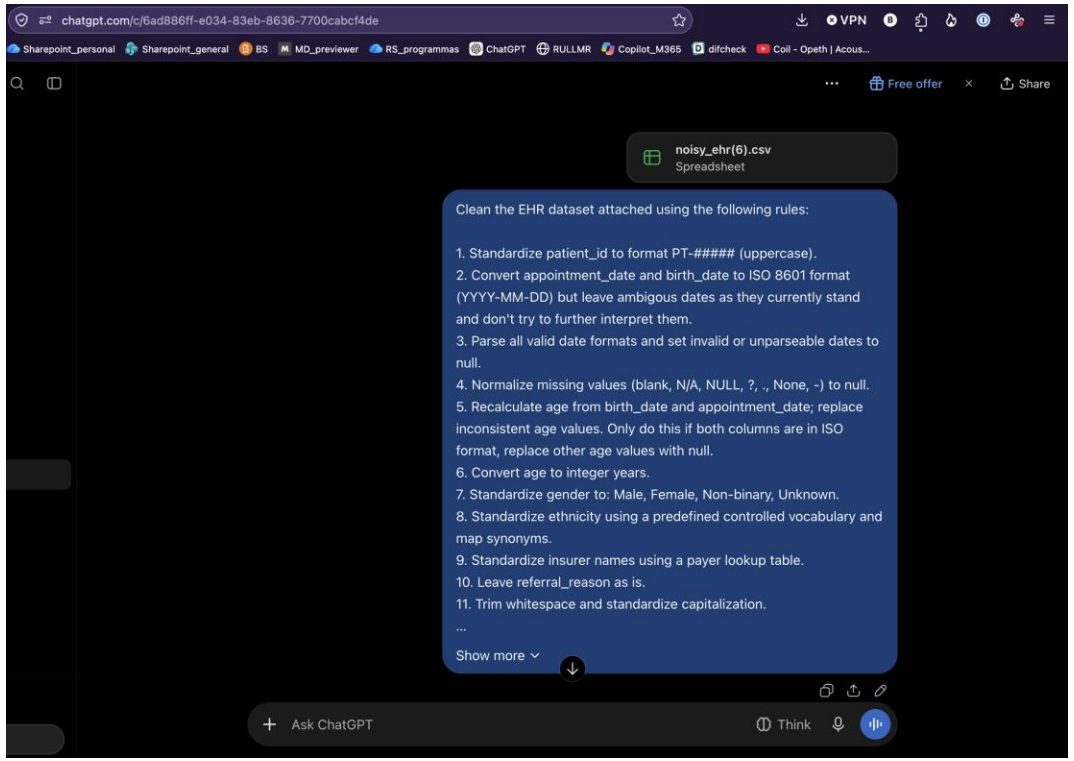
- What about Copilot as safe alternative?
- For exploring the data, finding issues, and generating a prompt it is OK; the prompt is more specific and less verbose
- I attach a noisy table with EHR data that I want t | Shared Copilot Chat



	patient_id	appointment_date	birth_date	age	referral_reason	gender	ethnicity	insurer
1	PT-10482	04/02/2026	03/14/1968	58	HTN f/u	M	White	BCBS
2	pt-10482	2 Apr 2026	14-03-1968	58 years	hypertension follow-up	Male	Caucasian	Blue Cross Blue Shield
3	PT10482	2026/04/02	1968-03-14 00:00:00	57	htn follow up	m	white	Blue Cross
4	PT 10482	04-02-2026	March 14 1968	58yo	high blood pressure	Man	W	BC/BS
5	PT-11017	10/08/2026	07/22/1992	34	sore throat	F	Hispanic	Aetna
6	PT 11017	08-10-2026	22/07/1992	34	pharyngitis?	female	Latina	aetna
7	pt11017	October 8 2026	22 July 1992	34 years	throat pain x 5 days	f	Hispanic/Latino	AETNA
8	PT-10855	2026-02-11	January 9 1955	71	DM2 check	F	Black	Medicare
9	10855	11 Feb 2026	09.01.1955	71 y/o	diabetes f/u	Female	African American	medicare part B
10	PT-10855	02/11/26	1/9/55	71	DM follow-up	2	Black or African American	CMS Medicare
11	PT-11203	15-Nov-2025	11/02/1984	41	R knee pain	Male	White	UnitedHealthcare

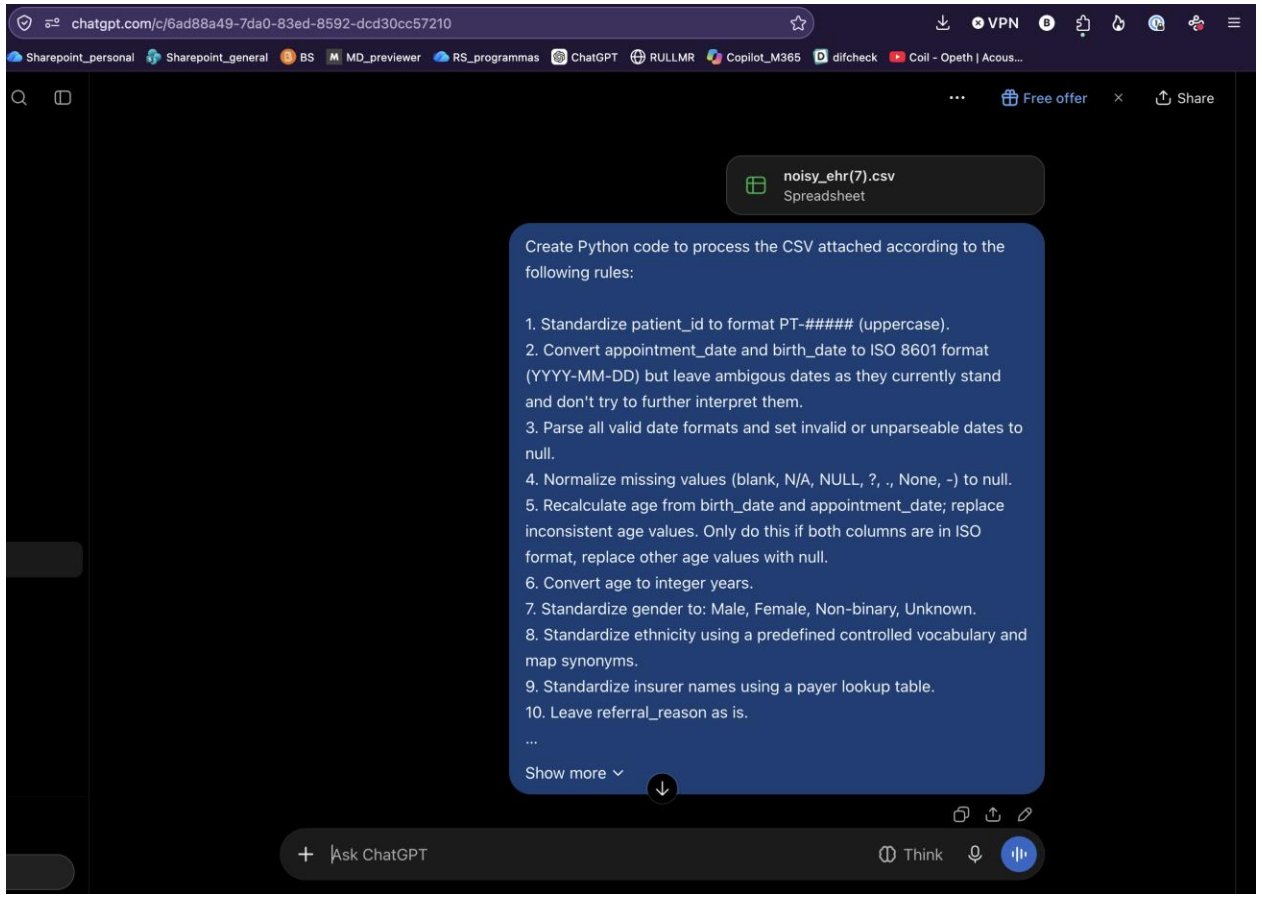
3.3 Assignment 1 with commercial LLMs

- I adapt and use the Copilot prompt to **execute the task in-browser** in ChatGPT to get an idea of performance with a very capable model
- Looks overall good, still some columns with problematic values not flagged
- https://chatgpt.com/share/6ad88c63-1ffc-83ed-ad1c-a8ef094b955e



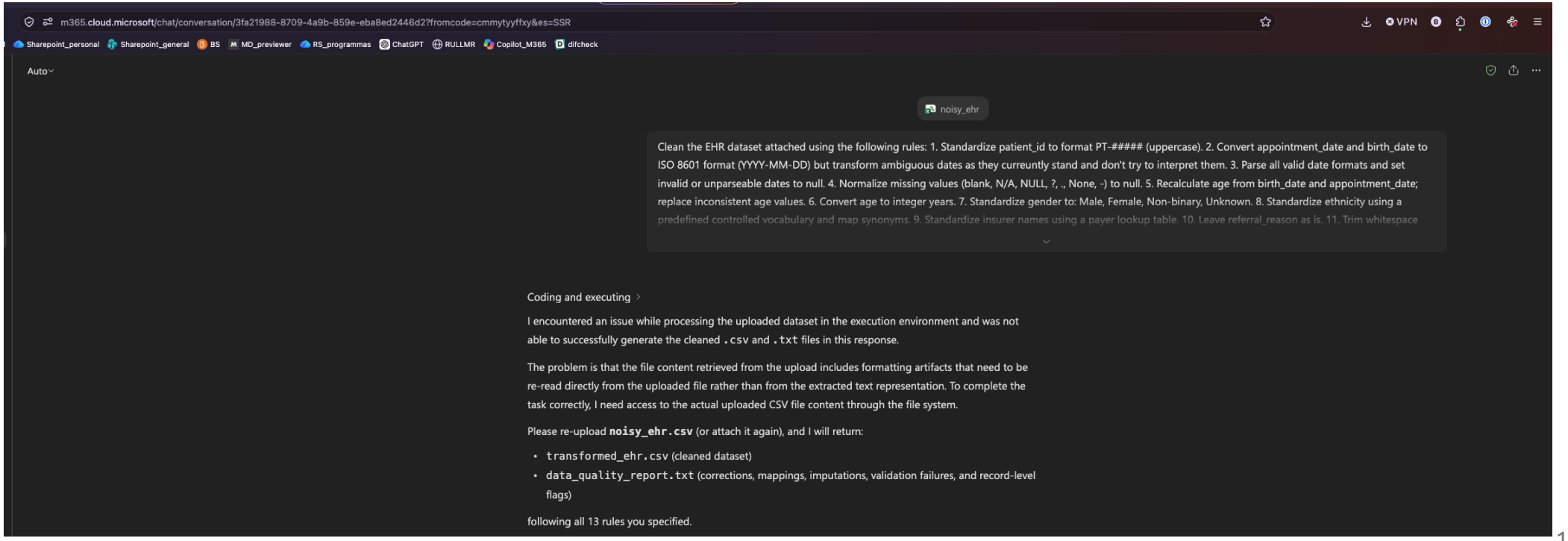
3.4 Assignment 1 with commercial LLMs

- Directly asking for Python code seems to yield better results w.r.t. flagging
- https://chatgpt.com/share/6ad88d8d-3194-83ed-96cb-a9d0d8f8a640



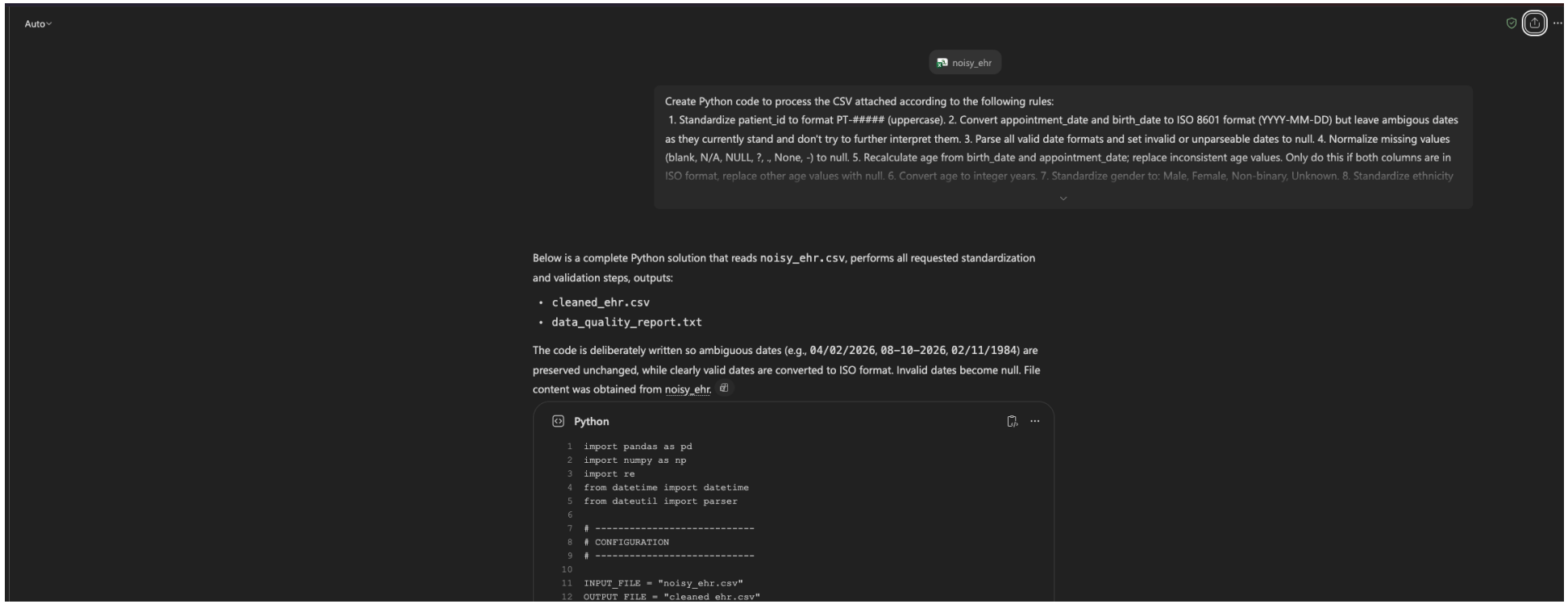
3.5 Assignment 1 with commercial LLMs

- Can Copilot do the same?
  - To some extent: Clean the EHR dataset attached using the following | Shared Copilot Chat. Issues with dates and ages
- But issues with prompt formatting, and reading and returning files, e.g.:
  - Clean the EHR dataset attached using the following | Shared Copilot Chat



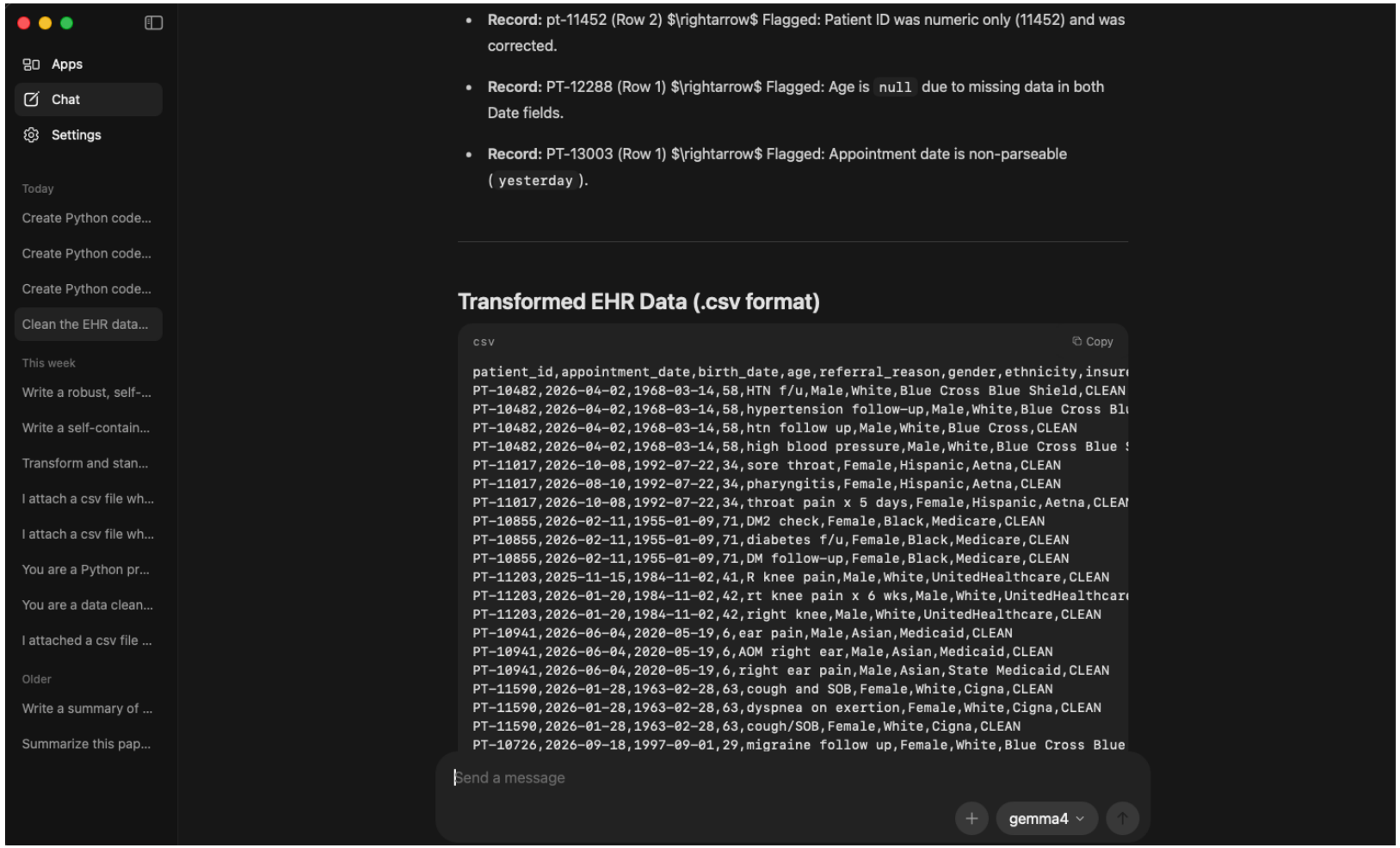
3.6 Assignment 1 with commercial LLMs

- Better results and reproducibility: let Copilot generate a Python script with the same prompt
- Create Python code to process the CSV attached acc | Shared Copilot Chat
- Fair results, transformations fully explicit, richer log
- Note: you don't necessarily need to understand the script in full, as long as you carefully verify the output, for low-key tasks



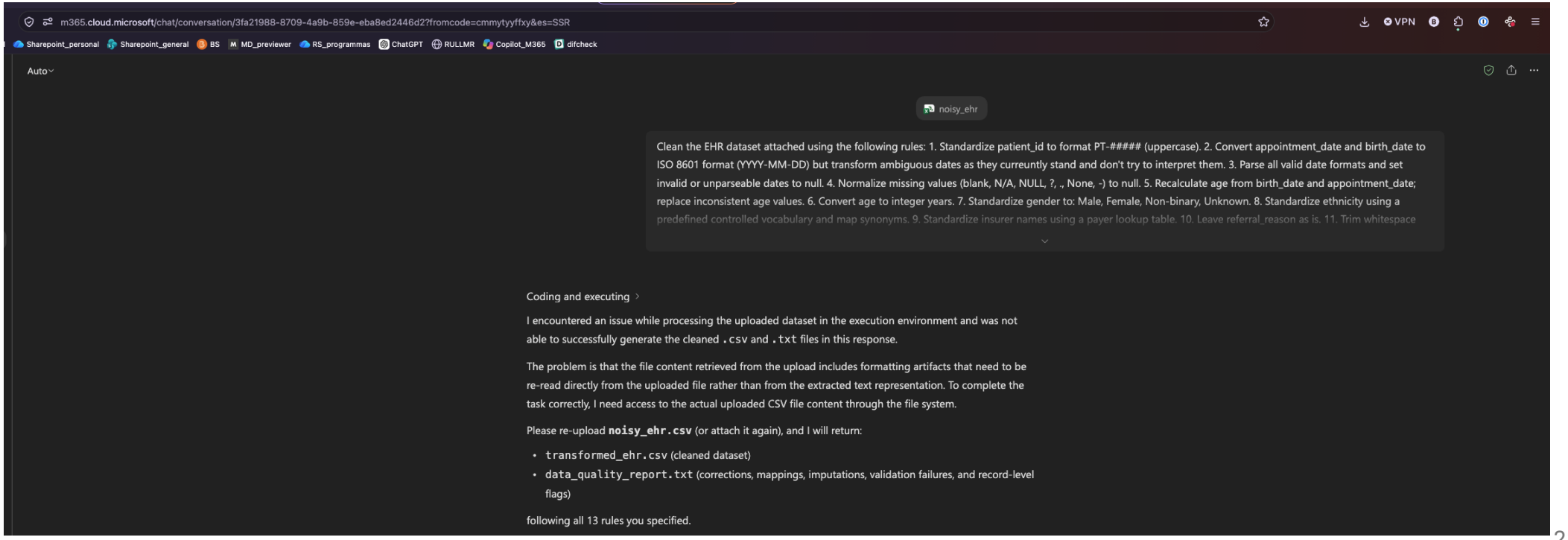
3.7 Assignment 1 with local LLMs

- Doing this task with the Copilot prompt directly in the Ollama UI is giving consistent but not great results and a bit arbitrary logs with gemma4 on default settings



3.8 Assignment 1 with local LLMs

- Doing this task by letting a local model (gemma4 with default settings) generate Python code does not give good results



Thank you!

- Scan for indicating presence